VESSEL TRAFFIC SERVICES (VTS)

Welcome to study Maritime English: *Vessel Traffic Services*. This module is divided into two parts. The first one is called *VTS in theory*. It explains the statutory background of Vessel Traffic Services. In this part the exercises concentrate on words and phrases which are typical of statutory texts. The other part is called *VTS in practice.* It consists of dialogues and exercises based on these dialogues.

It must be noted that the dialogues are imaginary although real ship names and photos are used in the exercises. The traffic situations described in the dialogues have been created for the purposes of the MarEng Project.

VTS, Text 1

What are vessel traffic services?

Growth in international commerce and tourism as well as technical development has resulted in high traffic density in many sea areas. Moreover, vessels are bigger and faster. The approaches to pilot boarding stations, ports and river waterways are all areas of potential traffic congestion and complexity. To reduce the risk of collision in busy port waters as well as in more open coastal waterways vessel traffic services (VTS) have been implemented. Vessel traffic services are shore-side information processing systems which range from the provision of simple information messages to ships, to extensive management of traffic within a port or waterway. Information messages can include position of other traffic, defects in aids to navigation or meteorological hazard warnings.
As stated in the IMO Resolution A.857(20), the efficiency of a VTS will depend on the reliability and continuity of communications and on the ability to provide good and unambiguous information. Furthermore, the quality of accident-prevention measures will depend on the system's capability of detecting a developing dangerous situation and on the ability to give timely warning of such dangers.

This text has been adapted from the following sources:

IMO Assembly Resolution A.857(20) paragraph 2.1.3

IMO Internet page: www.imo.org/Safety

Education and Culture
Leonardo da Vinci

VTS, Text 2

What is the purpose of vessel traffic services?

As stated in IMO Resolution A.857(20) the purpose of vessel traffic services is threefold: firstly, to improve the safety and efficiency of vessel traffic, secondly, to improve the safety of life at sea and, thirdly, to protect the marine environment and the adjacent shore area, worksites and offshore installations from possible adverse effects of maritime traffic. For example, in the event of pollution resulting from a collision or grounding, VTS helps to limit the effects by working with other shore-based agencies and directing other vessels to avoid the area. Also, a VTS has a valuable role in helping to identify the source of the pollution.

Vessel traffic services apply to all merchant and government vessels navigating in an area where these services are provided. Depending upon governing rules and regulations, participation in a VTS may be mandatory or voluntary. Participation by leisure craft is voluntary, but they, too, should follow the instructions given by the VTS Centre.

One important feature in the legal position of the VTS is the possibility of giving instructions to vessels when the provision of information has not lead to the desired result. This might happen if a vessel is not acting in accordance with the agreed procedures in an area covered by a VTS.

This text has been adapted from the following sources:

IMO Assembly Resolution A.857(20) paragraph 2.1.1

VTS Master's Guide. Gulf of Finland. Finnish Maritime Administration, Helsinki. Available at: http://www.fma.fi

VTS, Text 3

What are the services provided?

The benefits of implementing a VTS are summarised in IMO Resolution A.857(20). VTS allows identification and monitoring of vessels, strategic planning of vessel movements and provision of navigational information and assistance. It can also assist in prevention of pollution and co-ordination of pollution response.

IMO Resolution A.857(20) paragraph 2.1.2 says that "a clear distinction may need to be made between a Port and Harbour VTS and a Coastal VTS. A Port VTS is mainly concerned with vessel traffic to and from a port or harbour or harbours, while a Coastal VTS is mainly concerned with vessel traffic passing through the area. A VTS could also be a combination of both types. The type and level of service or services rendered could differ between both types of VTS; in a Port or Harbour VTS a navigational assistance service and/or a traffic organi[s]ation service is usually provided for, while in a Coastal VTS usually only an information service is rendered."

Source: IMO Assembly Resolution A.857(20) paragraph 2.1.2

VTS, Text 4

How do vessel traffic services function?

The main tool of a VTS operator is a traffic image. It is a comprehensive overview of the traffic in the area combined with all traffic influencing factors. Technically, a traffic image is a combination of information from different sources. First, the ships' movements are monitored by radar. Then, the computer combines the radar image with an electronic navigational chart which displays the fairway, its aids to navigation and the depth information. The VTS operators monitor vessel traffic on screens where ships' radar echoes are identified and put under surveillance. The operators can monitor ships' movements online, study their previous passage and predict their future path on the basis of the course and speed they hold.

A VTS operator has specialised knowledge of the waterway and has, therefore, responsibility for managing the traffic in the area. The master of a vessel has knowledge of the behaviour of the vessel. Therefore, responsibility for safe navigation lies with the ship's master at all times. Neither a VTS passage plan, nor requested or agreed changes to the passage plan, can supersede the decisions of the master concerning the actual navigation and manoeuvring of the vessel. Any instruction from a VTS to a vessel should be "result oriented" only, leaving the details of execution to the master, officer of the watch or pilot on board the vessel.

VTS, text 5

Automatic Identification System (AIS)

The implementation of the Automatic Identification System (AIS) has considerably enhanced safety at sea. AIS is an aid to monitoring vessel traffic. The system makes it possible to get information about ships and their movements at intervals of a few seconds. Today, all ships of 300 gross tonnage and upwards are required to be fitted with shipborne automatic identification systems. This requirement is met by installing a shipborne VHF transceiver which operates globally on two dedicated VHF channels. The shipborne AIS continuously and automatically transmits fixed, dynamic and voyage-related information and receives corresponding information from other ships. AIS can provide the following information for vessels within the radio range and for automatic display in the VTS Centre:

1) Position
2) Call sign and Name
3) IMO and MMSI number
4) Type of ship
5) Length and beam
6) Navigational status (Underway, at anchor, etc.)
7) Speed over ground (SOG)
8) Course over ground (COG)
9) Heading
10) Rate of turn (ROT)

The available voyage related information includes:

1) Type of cargo
2) Ship's draught
3) Destination
4) Estimated Time of Arrival (ETA)

There is also a capability for shore to ship, ship to shore and inter-ship transmission of text messages. For VTS, the Automatic Identification System substantially improves monitoring of vessel traffic compared with traditional radar systems. All transponder targets within VHF radio range are automatically displayed and identified on digital charts. The influence of bad weather and target swapping experienced in radar echoes does not exist with the AIS technology.

Source: *Automatic Identification System (AIS).* Available at the Finnish Maritime Administration web-page: http://www.fma.fi/e/functions/trafficmanagement

VTS, Text 6

TRAFFIC REPORT

According to IMO Resolution A.857(20) vessels should make all required reports, including reporting of deficiencies, prior to entering a VTS area. The VTS Authority operating the VTS is responsible for providing mariners with information about the times and geographical positions for submitting reports, radio frequencies to be used for reporting and the form and content of the reports.

Vessels heading for the ports of Kotka, Hamina or Loviisa in the eastern Gulf of Finland must give a traffic report to the Kotka VTS Centre via VHF when entering the Kotka VTS area. This applies to vessels with length of over 24 metres. When a ship contacts the VTS Centre, it should give its name, location, the planned route and planned anchoring (who, where, destination). The VTS Centre will confirm that it has received the report and will provide the vessel with the necessary information and instructions.

A vessel must also give a traffic report after anchoring or after it has been made fast to the quay. A vessel manoeuvring irregularly must always give a traffic report. A report must also be given whenever circumstances so require.

This text has been adapted from the following sources:

IMO Resolution A.857(20) paragraph 2.6.3

VTS Master's Guide, Gulf of Finland. Finnish Maritime Administration, Helsinki. Pages 24-25. Available at: http://www.fma.fi

VTS, Text 7

Information Service

information service
a service of VTS to ensure that essential information becomes available in time for on-board navigational decision-making (IMO Res. A.857(20) paragraph 1.1.9.1)

According to the definition given in IMO Resolution A.857(20) the aim of information service is to ensure that essential information becomes available in time for on-board navigational decision-making. The VTS Centre receives and provides information about conditions and events important to shipping and safety at sea. Priority is given to information that is of immediate concern to the vessels in the area. Also vessels in the VTS area are obliged to report to the VTS Centre any observations that could affect safety at sea.

The information service is provided by broadcasting information at fixed times and intervals or when deemed necessary by the VTS centre. Information can also be given to a particular vessel in conjunction with the vessel's position report, on request or whenever circumstances so require. Information may include, for example, reports on the position, identity and intentions of other traffic, waterway conditions, weather, hazards, or any other factors that may influence the vessel's transit.

This text has been adapted from the following sources:
IMO Assembly Resolution A.857(20) paragraph 1.1.9.1
IMO Assembly Resolution A.857(20) paragraph 2.3.1
VTS Master's Guide, Gulf of Finland. Finnish Maritime Administration, Helsinki. Page 21, 25.

VTS, Text 8

Navigational Assistance Service

navigational assistance service a service of VTS to assist on-board navigational decision-making and to monitor its effects (IMO Res. A.857(20) paragraph 1.1.9.2)

According to the IMO Resolution A.857(20) the aim of navigational assistance service is to assist on-board navigational decision-making and to monitor its effects. This service is provided only on specified occasions and under clearly defined circumstances. The navigational assistance service is especially important in difficult navigational or meteorological circumstances or in case of defects or deficiencies. This service is normally rendered at the request of a vessel or when deemed necessary by the VTS Centre. When navigational assistance service is provided the beginning and end of navigational assistance should be clearly stated and acknowledged.

Not all VTS centres, though, are authorised to provide navigational assistance service. For example, Helsinki VTS or Kotka VTS do not provide vessels with any direct navigational instructions.

This text has been adapted from the following sources:

IMO Resolution A.857(20) paragraph 1.1.9.2

IMO Resolution A.857(20) paragraph 2.3.2

VTS, Text 9

Traffic Organisation Service

traffic organisation service a service of VTS to prevent the development of dangerous maritime traffic situations and to provide for the safe and efficient movement of vessel traffic within the VTS Area (IMO Res. A.857(20) paragraph 1.1.9.3)

According to the IMO Resolution A.857(20) the aim of traffic organisation service is to prevent the development of dangerous maritime traffic situations and to provide for the safe and efficient movement of vessel traffic within the VTS area. The traffic organisation service concerns the operational management of traffic and the forward planning of vessel movements to prevent congestion and dangerous situations. Operational management of traffic includes allocation of space, mandatory reporting of movements in the VTS area, routes to be followed, speed limits to be observed or other appropriate measures which are considered necessary by the VTS authority. The service is particularly relevant in times of high traffic density or when the movement of special transport may affect the flow of other traffic.

When necessary VTS gives instructions on the speed of vessels, prohibits vessels from overtaking other ships in the VTS area or specifies the right-of-way in narrow channels. The service may also include establishing and operating a system of traffic clearances or VTS sailing plans or both. A sailing plan is a mutually agreed plan between a VTS Authority and the master of a vessel and concerns the movement of the vessel in a VTS area.

This text has been adapted from the following sources:
IMO Resolution A.857(20) paragraph 1.1.9.3
IMO Resolution A.857(20) paragraph 2.3.3
VTS Master's Guide, Gulf of Finland. Finnish Maritime Administration, Helsinki. Pages 21-22, 25.

VTS, Dialogue 1

Entering report

Situation description:

The M/S Marina is entering the Kotka VTS area and gives a traffic report on VHF channel 67. The Kotka VTS centre confirms that it has received the report and provides the M/S Marina with necessary information and instructions.

Ship:	Kotka VTS, this is Marina.
VTS:	Marina, Kotka VTS.
Ship:	We are passing the reporting point number 10. My ETA at pilot station is 1300 hours local time.
VTS:	Marina, Kotka VTS. You are in the Kotka VTS area. Proceed to Orrengrund Pilot Station. **Information:** Rig the pilot ladder on starboard side, one metre above the water. Make a boarding speed of six knots.
Ship:	Pilot ladder starboard side, one metre above the water. Boarding speed six knots. Is pilot ready for me?
VTS:	Yes, pilot on arrival. **Traffic information:** One outbound vessel, name Annegrecht, now passing Tainio light.
Ship:	Okay, traffic information received.
VTS:	**Information:** There are cable operations in position 277 degrees from the southern point of Kaunissaari island distance 4 miles. Wide berth requested.
Ship:	Cable operations in vicinity of Kaunissaari. Wide berth requested. Well received.

VTS, Dialogue 2

Information Service

Situation description:

The M/S Marina has entered the Kotka VTS area and is heading for Kotka, the port of destination. The Kotka VTS centre notices that Marina is proceeding to deep water fairway pilot boarding position which is west of Kotka lighthouse. However, deep water pilot boarding position is for vessels with the draught of 10 metres or more. Marina's present draught is 7.5 metres. The Kotka VTS centre informs Marina that the pilot is boarding at the Orrengrund pilot boarding position, 1.5 SSW of Orrengrund. Now Marina is proceeding towards Tainio racon.

VTS: Marina, Kotka VTS.
Ship: Kotka VTS, Marina.

VTS: Marina, Kotka VTS. **Information:** The fairway is on the west side of Tainio racon.
Ship: Yes. I will alter course to port.

The ship is proceeding towards Tainio racon.

VTS: Marina, Kotka VTS. **Warning:** You are running into danger. The fairway is on the west side of Tainio racon.
Ship: Okay, I will leave it on my starboard side.

VTS: **Information:** One outbound vessel, name Finnmill, passing Orrengrund after ten minutes.
Ship: Understood, Finnmill is coming out. I can see her on my radar screen.

VTS: **Information:** There is a yacht race on the west side of Kotka lighthouse. The yachts are heading north-west.
Ship: Well understood.

VTS, Dialogue 3

Navigational Assistance Service

Situation description:

The M/S MSC Marianna is in the Hightower VTS area. Suddenly the mate notices that only one radar is working. Visibility is 500 metres. He decides to request navigational assistance. The Hightower VTS starts navigational assistance and informs other vessels in the area.

Ship: Hightower VTS, this is MSC Marianna.
VTS: MSC Marianna, Hightower VTS.

Ship: **Information:** We have problems with electricity on the bridge. Only one radar is working. I require navigational assistance.
VTS: Understood. You have problems with electricity. **Question:** What is your position?

Ship: **Answer:** My position is bearing 035 degrees, distance 5.5 miles from Landfall lighthouse.
VTS: Bearing 035 degrees, distance 5.5 miles from Landfall lighthouse. I have located you on my radar screen. All information is based on VTS equipment. Stand by on channel 10. If you do not hear from me in one minute time, navigational assistance is ended. In that case go back to channel 71 and call Hightower VTS. Navigational assistance starts at 0920 local time.

Ship: Okay. Navigational assistance is starting and is provided on channel 10. If I do not hear from you at one minute intervals, navigational assistance is ended and I will call Hightower VTS on channel 71.
VTS: All ships, Hightower VTS. **Traffic information:** Container vessel MSC Marianna, just passed buoy number four, is receiving navigational assistance on channel 10 and is not monitoring any other VTS working channel.

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VTS: You are on the centre of the fairway, tendency to north. Bearing to the next buoy is 120 degrees, distance 2.2 miles.

Ship: Thank you. Information: My next waypoint is in position 240 degrees and 0.2 miles from buoy number six. After that I will commence the turn to the next course which is 090.

VTS: Next waypoint in position 240 degrees and 0.2 miles from buoy number six. Understood.

...

VTS: You are on the northern buoy-line of the fairway, tendency to north. Bearing to the next waypoint is 140 degrees, distance 1.8 miles.

Ship: Understood. I am on the northern buoy-line. Bearing to the next waypoint 140 degrees. I will alter my course to south.

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Ship: I am passing buoy number two and my navigational equipment is working now. I no longer require navigational assistance. Thank you.

VTS: MSC Marianna, Hightower VTS. Information received. Navigational assistance ends at 0945 local time.

VTS, Dialogue 4

Traffic Organisation Service

In the Helsinki or Kotka VTS area a vessel has to obtain permission from the VTS Centre to depart not more than five minutes before casting off from the quay or leaving the anchorage and must report its planned route. After checking the traffic situation, the VTS Centre will notify the vessel of other ships in the area and, if necessary, will specify the order of departure. A ship must notify the VTS Centre of any inability to keep to the planned departure time. If necessary, the VTS Centre will recommend an alternative route to the vessel.

Situation description:

The following happens in the Kotka VTS area. The M/S Janra gives a traffic report stating that she has left Halla lo-lo berth bound for Orrengrund pilot station and will be in the Strait of Ruotsinsalmi after 10 minutes. The M/S Greta is leaving Sunila quay bound for Halla via the Strait of Ruotsinsalmi.

Ship: Kotka VTS this is Janra.
VTS: Janra, Kotka VTS.

Ship: **Information:** I am ready to depart from Halla and I am bound for Orrengrund.
VTS: Traffic clearance. Janra has permission to depart from Halla to Orrengrund at 0900 local time.

Ship: We have permission to depart.
VTS: **Traffic information:** One outbound vessel Ladoga-53 has just passed buoy number 2 and tanker Berg is anchored in position one mile south from Vehkaluoto.
Ship: Okay, thank you for your information. I can see Ladoga-53.

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Ship: Kotka VTS, Janra. We have 10 minutes to the Strait of Ruotsinsalmi.
VTS: Janra, Kotka VTS. You have 10 minutes to the Strait of Ruotsinsalmi. There is no other reported traffic in the strait.

Ship: Well received.

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Ship: Kotka VTS, this is Greta. I request clearance to depart from Sunila bound for Halla. I am just shifting berth.

VTS: Greta, Kotka VTS. Negative. Do not proceed. **Traffic information:** Container vessel Janra has two minutes to the Strait of Ruotsinsalmi. I will contact you when she has passed the strait and the fairway is free.

Ship: Well received. No permission to depart. I will remain alongside.

VTS, Dialogue 5

Engine problem

Ship: Kotka VTS, this is Marina.

VTS: Marina, Kotka VTS.

Ship: Kotka VTS, Marina. We have engine problems and I am manoeuvring with difficulty.

VTS: Marina, Kotka VTS. Do you need assistance?

Ship: Negative, no assistance needed.

VTS: Understood. Keep me updated about your situation.

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Ship: Kotka VTS, this is Nordic Star.

VTS: Nordic Star, Kotka VTS.

Ship: **Intention:** Nordic Star is ready to depart from berth number C5 to sea. Can I have clearance now?

VTS: Negative. Do not proceed. **Traffic information:** Tanker Paula is just departing from oil terminal. Call VTS again after Paula has passed you.

Ship: Well understood. We do not have permission to depart. We will stay alongside. I will call you again when tanker Paula has passed me.

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VTS: Friedrich Russ, Kotka VTS.

Ship: Kotka VTS, this is Friedrich Russ.

VTS: **Information:** Outbound vessel Marina in the vicinity of buoy number 10 has engine problems and she is manoeuvring with difficulty. **Advice**: Keep minimum passing distance of 5 cables when passing her.

Ship: Well understood. I will keep clear of Marina.

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Ship: Kotka VTS, Marina.
VTS: Marina, Kotka VTS.

Ship: My engines are working now and I can make a speed of 5 knots. I am still manoeuvring with difficulty.
VTS: Understood. **Traffic information:** Outbound vessel Friedrich Russ has just passed buoy number 8.
Ship: Friedrich Russ has passed buoy number 8 outbound. Okay. I will keep to the east side of the fairway.

VTS: Friedrich Russ, Kotka VTS.
Ship: Kotka VTS, Friedrich Russ. I heard that Marina is keeping to the east side of the fairway. I am passing Marina on her west side maintaining the minimum passing distance of 5 cables.
VTS: Friedrich Russ, Kotka VTS. Thank you.

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Ship: Kotka VTS, Marina.
VTS: Marina, Kotka VTS.

Ship: I still have problems with engines. **Question:** Where can I anchor?
VTS: **Question:** What is your maximum draught?
Ship: **Answer:** My maximum draught is 6.5 metres.
VTS: **Information:** You must anchor in position 1.5 miles south from Tainio racon.
Ship: Okay, I will anchor at given position.

VTS: Call Kotka VTS after you have anchored.
Ship: Okay, I will call Kotka VTS after I have anchored.